CHE 201 Spring 2020 HW 5

Thanh Nguyen

TOTAL POINTS

95 / 100

QUESTION 1

1 Question 1 15 / 15

✓ **- 0 pts** Correct

- 5 pts wrong spercific volume

- 5 pts wrong T2

- 5 pts wrong T when the tank contains only saturated vapor

- 15 pts Question not answered

QUESTION 2

2 Question 2 10 / 15

- 0 pts Correct

- 5 pts wrong v1

- 5 pts wrong v2

✓ **- 5 pts** wrong work

- 15 pts Question not answered

QUESTION 3

3 Question 3 15 / 15

✓ **- 0 pts** Correct

- 5 pts Wrong T-v diagram

- 2.5 pts wrong U1

- 2.5 pts wrong u2

- 2.5 pts wrong electricity work

- 2.5 pts wrong time

- 15 pts Question not answered

QUESTION 4

4 Question 4 20 / 20

✓ **- 0 pts** Correct

- 20 pts Question not answered

- 5 pts wrong u1

- 5 pts wrong u2

- 5 pts wrong electricity work

- 5 pts wrong final heat transfer

QUESTION 5

5 Question 5 20 / 20

✓ **- 0 pts** Correct

- 5 pts wrong sketch of p-v diagram

- 2.5 pts wrong Q1

- 2.5 pts wrong W1

- 2.5 pts wrong Q2

- 2.5 pts wrong W2

- 2.5 pts wrong Q3

- 2.5 pts wrong W3

- 20 pts Question not answered

QUESTION 6

6 Question 6 15 / 15

✓ **- 0 pts** Correct

- 2.5 pts not enough components of ways to improve water use

- 5 pts not enough components of ways to improve water use

- 7.5 pts not enough components of ways to improve water use

- 2.5 pts not enough components of record of water use and compare

- 5 pts not enough components of record of water use and compare

- 15 pts Question not answered

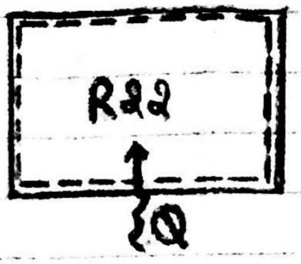
Thanh Nguyen

CHE 201 – HW5

1.) <u>Sketch</u>

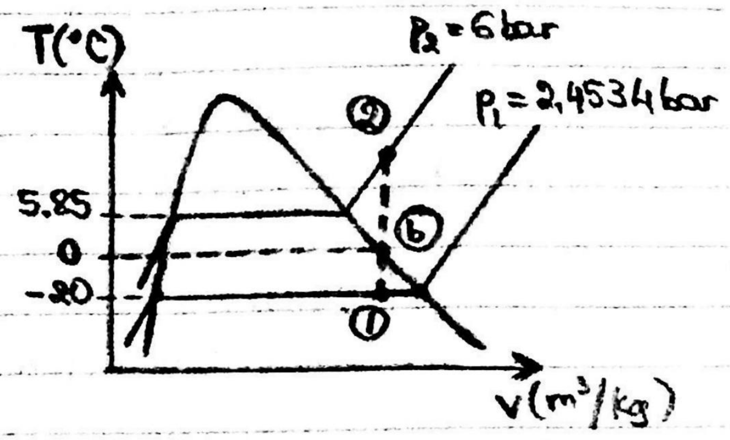
Initial state

- Liquid-vapor mixture
- $T_1 = -20°C$; v_1 ; p_1
- $X = 0.5036$



Final state

- $p_2 = 6$ bar ; T_2 ; v_2



<u>Engineering Model</u>

1. The system is closed
2. There is no change in volume
   → constant specific-volume process

<u>Analysis</u>

(a) From table A-7, at $T_1 = -20°C \rightarrow v_{f1} = 0.7427 \times 10^{-3} m^3/kg$; $v_{g1} = 0.0926 m^3/kg$ and $p_1 = 2.4534$ bar

$\rightarrow v_1 = (1-x)v_{f1} + x v_{g1} = (1-0.5036)(0.7427 \times 10^{-3} m^3/kg) + (0.5036)(0.0926 m^3/kg)$
$= 0.047 m^3/kg$

$\rightarrow v_2 = v_1 = 0.047 m^3/kg$

From table A-8, at $p_2 = 6$ bar $\rightarrow v_{f2} = 0.7927 \times 10^{-3} m^3/kg$; $v_{g2} = 0.0392 m^3/kg$; $T_2 = 5.85°C$

Since $v_2 > v_{g2}$, state 2 is in the superheated vapor region.

From table A-9, at $p_2 = 6$ bar

- $T_a = 40°C \rightarrow v_a = 0.04628 m^3/kg$
- $T_b = 45°C \rightarrow v_b = 0.04724 m^3/kg$

By interpolation: $\frac{T_2 - T_a}{v_2 - v_a} = \frac{T_b - T_a}{v_b - v_a} \rightarrow \frac{T_2 - 40°C}{(0.047 - 0.04628) m^3/kg} = \frac{(45-40)°C}{(0.04724 - 0.04628) m^3/kg}$

$\rightarrow T_2 = 43.75°C$

(b) From table A-7, at $T = 0°C$ and $p = 4.9811$ bar, the tank will contain only saturated vapor with specific volume $v = v_1 = 0.047 m^3/kg$.

1 Question 1 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** wrong spercific volume
- **- 5 pts** wrong T2
- **- 5 pts** wrong T when the tank contains only saturated vapor
- **- 15 pts** Question not answered

2.) Sketch

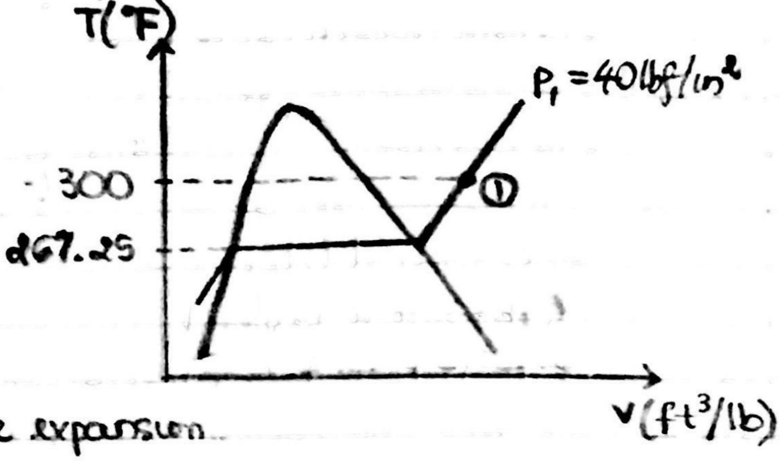
Initial state

$p_1 = 40\,lbf/in^2$

$T_1 = 300°F$

Final state

$p_2 = 14.7\,lbf/in^2$



Engineering Model

1. The system is closed
2. $pv^{1.2} = \text{constant}$
3. $\Delta KE = \Delta PE = Q = 0$
4. Work is only in the form of volume expansion

Analysis

From table A-3E, at $p_1 = 40\,lbf/in^2$, $T_{sat} = 267.25°F$

Since $T_1 > T_{sat}$, state 1 is in superheated vapor region.

From table A-4E, at $p_1 = 40\,lbf/in^2$ and $T_1 = 300°F$, $v_1 = 11.04\,ft^3/lb$

$$p_1 v_1^{1.2} = p_2 v_2^{1.2} \rightarrow v_2 = \left(\frac{p_1}{p_2}\right)^{1/1.2} \cdot v_1 = \left(\frac{40\,lbf/in^2}{14.7\,lbf/in^2}\right)^{1/1.2} (11.04\,ft^3/lb) = 25.42\,ft^3/lb$$

$$W = m\int_{v_1}^{v_2} p\,dv = m\int_{v_1}^{v_2} \frac{\text{constant}}{v^{1.2}}\,dv = m\,\frac{p_2 v_2 - p_1 v_1}{1-1.2}$$

$$= (0.5\,lb)\,\frac{(14.7\,lbf/in^2)(25.42\,ft^3/lb) - (40\,lbf/in^2)(11.04\,ft^3/lb)}{-0.4}$$

$$= 84.82\,lbf.ft^3/in^2 \times \left(\frac{12\,in}{1\,ft}\right)^2 \times \frac{1\,Btu}{778.2\,ft.lbf} = 15.70\,Btu$$

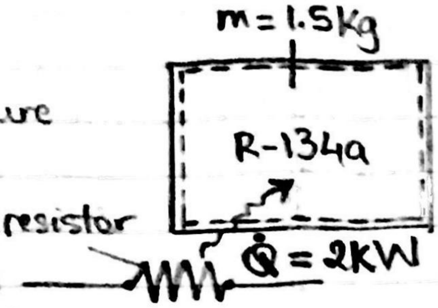
2 Question 2 **10 / 15**

- **- 0 pts** Correct
- **- 5 pts** wrong v1
- **- 5 pts** wrong v2
- ✓ **- 5 pts** wrong work
- **- 15 pts** Question not answered

3.) Sketch

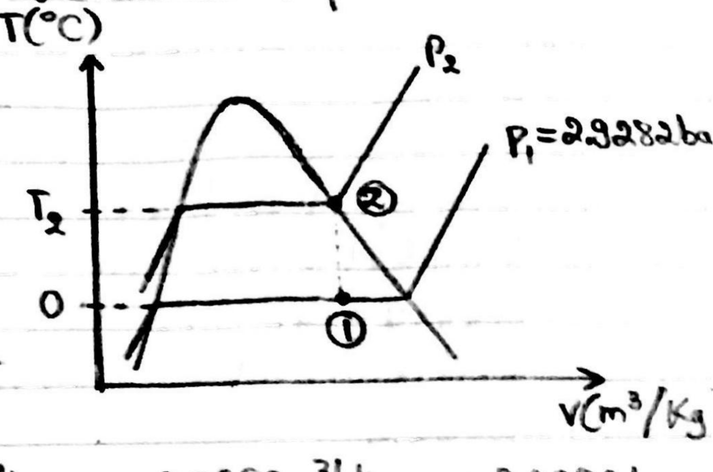
Initial state

- Liquid-vapor mixture
- $T_1 = 0°C$
- $x = 0.6$



Final state

- Saturated vapor



Engineering Model

1. The system is closed 2. $\Delta KE = \Delta PE = W = 0$
3. Constant specific volume: $v_1 = v_2$
4. v_g directly proportional to u_g over small ΔT

Analysis

From table A-10, at $T_1 = 0°C$, $v_{f1} = 0.7721 \times 10^{-3} m^3/kg$; $v_{g1} = 0.0689 m^3/kg$, $P_1 = 2.9282 bar$

$u_{f1} = 49.79 kJ/kg$; $u_{g1} = 227.06 kJ/kg$

$$u_1 = (1-x)u_{f1} + x u_{g1} = (1-0.6)(49.79 kJ/kg) + (0.6)(227.06 kJ/kg)$$
$$= 156.152 kJ/kg$$
$$v_1 = (1-x)v_{f1} + x v_{g1} = (1-0.6)(0.7721 \times 10^{-3} m^3/kg) + (0.6)(0.0689 m^3/kg)$$
$$= 0.04165 m^3/kg \rightarrow v_1 = v_2 = 0.04165 m^3/Kg$$

From table A-10

- At $T_a = 12°C$, $v_{ga} = 0.0460 m^3/kg$, $u_{ga} = 233.63 kJ/kg$
- At $T_b = 16°C$, $v_{gb} = 0.0405 m^3/kg$, $u_{gb} = 235.78 kJ/kg$

By interpolation: $\dfrac{u_{ga} - u_{gb}}{v_{ga} - v_{gb}} = \dfrac{u_2 - u_{ga}}{v_2 - v_{ga}} \rightarrow \dfrac{(233.63 - 235.78) kJ/kg}{(0.0460 - 0.0405) m^3/kg} = \dfrac{u_2 - 233.63 kJ/kg}{(0.04165 - 0.0460) m^3/kg}$

$$\rightarrow u_2 = 235.33 kJ/Kg$$
$$\Delta E = Q - W \rightarrow \Delta U = Q \rightarrow m\Delta u = \dot{Q}\Delta t$$
$$\rightarrow \Delta t = \frac{m(u_2 - u_1)}{\dot{Q}} = \frac{(1.5 kg)(235.33 kJ/kg - 156.152 kJ/kg)}{2kW} = 59.38 s$$

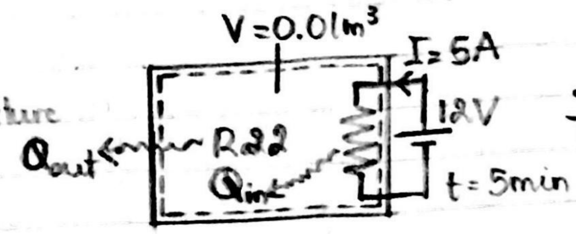
3 Question 3 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** Wrong T-v diagram
- **- 2.5 pts** wrong U1
- **- 2.5 pts** wrong u2
- **- 2.5 pts** wrong electricity work
- **- 2.5 pts** wrong time
- **- 15 pts** Question not answered

4.) Sketch

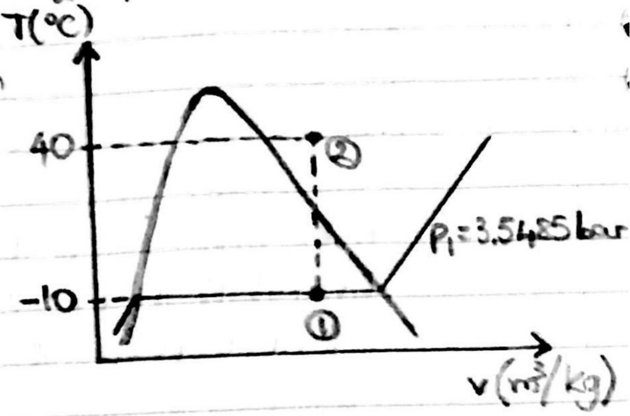
Initial state

- Liquid-vapor mixture
- $T_1 = -10°C$
- $x_1 = 0.8$



Final state

- $T_2 = 40°C$



Engineering Model

1. The system is closed
2. Constant specific volume $v_1 = v_2$
3. $\Delta KE = \Delta PE = W = 0$
4. No heat is lost along the wire : $\dot{Q}_{in} = P_{resistor}$

Analysis

From table A-7, at $T_1 = -10°C$, $v_{f_1} = 0.7606 \times 10^{-3} m^3/kg$; $v_{g_1} = 0.0652 m^3/kg$; $P_1 = 3.5485$ bar

$u_{f_1} = 33.27 kJ/kg$, $u_{g_1} = 223.02 kJ/kg$

$v_1 = (1-x_1)v_{f_1} + x v_{g_1} = (1-0.8)(0.7606\times10^{-3} m^3/kg) + (0.8)(0.0652 m^3/kg)$

$= 0.05231 m^3/kg \rightarrow v_1 = v_2 = 0.05231 m^3/kg$

$u_1 = (1-x_1)u_{f_1} + x u_{g_1} = (1-0.8)(33.27 kJ/kg) + (0.8)(223.02 kJ/kg) = 185.07 kJ/kg$

From table A-7, at $T_2 = 40°C$, $v_{f_{40}} = 0.8858 \times 10^{-3} m^3/kg$, $v_{g_{40}} = 0.0151 m^3/kg$

Since $V_2 > V_{g_{40}}$, state 2 is in superheated vapor region

From table A-9, at $T_2 = 40°C$

- $P_a = 5 bar \rightarrow V_a = 0.05636 m^3/kg$, $u_a = 250.70 kJ/kg$
- $P_b = 5.5 bar \rightarrow v_b = 0.05086 m^3/kg$, $u_b = 250.20 kJ/kg$

By interpolation $\dfrac{u_a - u_b}{V_a - V_b} = \dfrac{u_2 - u_b}{V_2 - V_b} \rightarrow \dfrac{(250.70-250.20) kJ/kg}{(0.05636-0.05086) m^3/kg} = \dfrac{u_2 - 250.20 kJ/kg}{(0.05231-0.05086) m^3/kg}$

$\rightarrow u_2 = 250.33 kJ/kg$

$Q_{in} = P_{resistor}\,\Delta t = V.I.\Delta t = (12V)(5A)(5)(60s) = 18000J = 18kJ$

$\Delta E = Q - W \rightarrow \Delta U = Q \rightarrow m\Delta u = Q_{in} - Q_{out}$

$\rightarrow Q_{out} = Q_{in} - \dfrac{V}{v_1}(u_2 - u_1) = 18kJ - \dfrac{0.01 m^3}{0.05231 m^3/kg}(250.33 kJ/kg - 185.07 kJ/kg)$

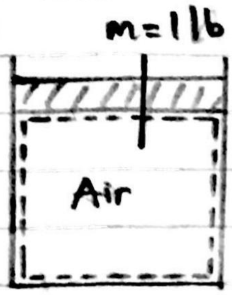
$= 5.52 kJ$

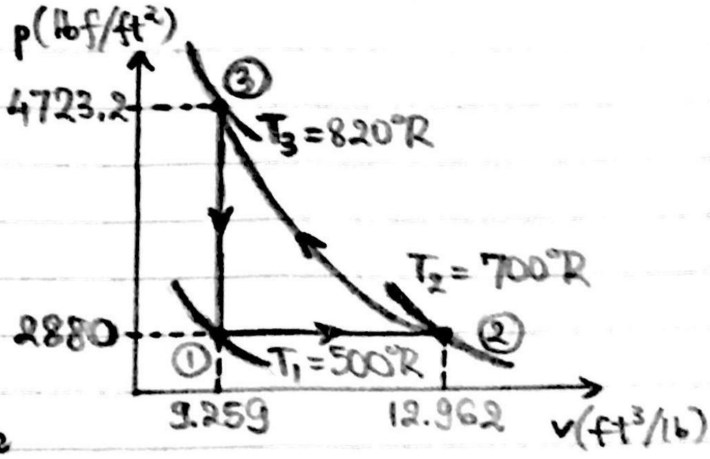
4 Question 4 **20 / 20**

- ✓ **- 0 pts** Correct
- **- 20 pts** Question not answered
- **- 5 pts** wrong u1
- **- 5 pts** wrong u2
- **- 5 pts** wrong electricity work
- **- 5 pts** wrong final heat transfer

5.) Sketch

- Process 1-2: $p_1 = p_2 = 20\,lbf/in^2$
  $T_1 = 500°R$, $v_2 = 1.4 v_1$
- Process 2-3: $Q_{23} = 0$
  $v_3 = v_1$, $T_3 = 820°R$
- Process 3-1: Constant volume





Engineering Model

1. The system is closed
2. Work is only in the form of volume change
3. $\Delta KE = \Delta PE = 0$
4. Ideal gas behavior

Analysis: $\Delta E = Q - W \rightarrow m\Delta u = Q - W$

- Process 1-2: $\frac{T_1}{T_2} = \frac{v_1}{v_2} \rightarrow T_2 = \frac{T_1 v_2}{v_1} = 1.4T_1 = 1.4(500°R) = 700°R$

From table A-22E, at $T_1 = 500°R \rightarrow u_1 = 85.20\,Btu/lb$; at $T_2 = 700°R \rightarrow u_2 = 119.58\,Btu/lb$

$p_1 = p_2 = (20\,lbf/in^2) \times \left(\frac{12in}{1ft}\right)^2 = 2880\,lbf/ft^2$

$p_1 v_1 = RT_1 = \frac{\bar{R}}{M}T_1 \rightarrow v_1 = \frac{\bar{R}T_1}{Mp_1} = \frac{(1545\,ft.lbf/lbmol.°R)(500°R)}{(28.97\,lb/lbmol)(2880\,lbf/ft^2)} = 9.259\,ft^3/lb$

$\rightarrow v_2 = 1.4v_1 = 12.962\,ft^3/lb$

$W_{12} = mp_1\Delta v = mp_1(v_2 - v_1) = (1lb)(2880\,lbf/ft^2)(12.962\,ft^3/lb - 9.259\,ft^3/lb) = 10664.64\,ft.lbf$

$\rightarrow W_{12} = 10664.64\,ft.lbf \times \frac{1\,Btu}{778.2\,ft.lbf} = 13.70\,Btu$

$\rightarrow Q_{12} = m(u_2 - u_1) + W = (1lb)(119.58\,Btu/lb - 85.20\,Btu/lb) + 13.70\,Btu = 48.08\,Btu$

- Process 2-3: $\frac{p_2 v_2}{T_2} = \frac{p_3 v_3}{T_3} \rightarrow p_3 = p_2 \frac{v_2}{v_3}\frac{T_3}{T_2} = (2880\,lbf/ft^2)(1.4)\frac{820°R}{700°R} = 4723.2\,lbf/ft^2$

From table A-22E, at $T_3 = 820°R \rightarrow u_3 = 140.47\,Btu/lb$

$Q_{23} = 0 \rightarrow W_{23} = -m\Delta u = -m(u_3 - u_2) = -(1lb)(140.47\,Btu/lb - 119.58\,Btu/lb)$
$= -20.89\,Btu$

- Process 3-1: $W_{31} = 0$

$\rightarrow Q_{31} = m\Delta u = m(u_1 - u_3) = (1lb)(85.20\,Btu/lb - 140.47\,Btu/lb) = -55.27\,Btu$

5 Question 5 **20 / 20**

- ✓ **- 0 pts** Correct
- **- 5 pts** wrong sketch of p-v diagram
- **- 2.5 pts** wrong Q1
- **- 2.5 pts** wrong W1
- **- 2.5 pts** wrong Q2
- **- 2.5 pts** wrong W2
- **- 2.5 pts** wrong Q3
- **- 2.5 pts** wrong W3
- **- 20 pts** Question not answered

Ways to Improve Water Use as a Society in Industry, Businesses, and Households

Ways to Improve Water Use in Industry

According to the United States Environmental Protection Agency (EPA), facility managers can find the following list a good way to reduce water use in industries:

- General management practices: designating a water efficiency coordinator, educating and involving employees in water efficiency plans and efforts.
- Equipment changes: installing high-efficiency toilets or water-saving devices on existing ones, installing high-pressure, low-volume nozzles and cleaning systems.
- Operating and maintenance procedures: detecting and repairing all leaks, using fogging nozzles to cool products, turning off all flows during shutdown, adjusting flowing in sprays and other lines to meet minimum requirements.
- Landscape irrigation: using properly treated wastewater for irrigation where available, watering the lawn or garden during the coolest part of the day, setting sprinklers to water the lawn or garden only (not the street or sidewalk), installing moisture sensors on sprinkler systems.

Ways to Improve Water Use in Businesses

The following list is suggested by Morgan Erwin of the Pacific Gas and Energy Company and EPA:

- Improving water system assessment and maintenance: developing a water-conservation plan by working with employees and educating them about water conservation, regularly assessing water-consuming systems to check for leaks.
- Implement simple, everyday water conservation techniques: checking for leaks, installing efficiently operating water systems (aerated faucets, water heater insulation, toilets, and so on), shutting off cooling units.
- Water recycling and water reuse: reusing boiler and steam condensate, using leftover drinking water to irrigate indoor plants and using wastewater to irrigate outdoor green areas.
- Using efficient landscaping techniques: installing moisture sensors, irrigating in the morning or early evening, checking sprinkler coverage areas.

Ways to Improve Water Use in Households

In order to reducing water consumption at home, Carol Crotta of Forbes suggests the following actions:

- Start saving water by breaking a bad habit: turning off faucets when it is unnecessary.
- Double-dip dishes: Instead of letting the water run who;e washing dishes, fill one sink with hot, soapy water for washing and the other with cool, clear water for rinsing, which will help reduce half the water usage for hand-dishwashing.

- Buy high-efficiency water systems: small dishwashers (if not hand-dishwashing), washers, toilets, and shower heads.
- Shorten the shower time: aiming for five minutes or less.
- Covering up the pool will retain the pool's temperature and reduce evaporation.
- Install new drip irrigation piping and soaker hoses for improved watering efficiency. Water by hand if having a small garden area.
- Capture rainwater for the use in the garden: remember to cover the barrels to keep mosquitoes away.

Comparing Daily Water Use

My normal daily water use includes:

- Toilet: 3 gallons per flush x 5-7 flushes per day = 15-21 gallons.
- Shower: 5-minute shower x 2 gallons a minute = 10 gallons.
- Hand-dishwashing: 3-7 gallons.
- Hygiene: 2 gallons.
- Drinking water: 1 gallon.
- Cooking: 3-10 gallons.
- Washing machine (one use per week): 10-15 gallons.

Total water use of my daily life is about 36-51 gallons. These numbers are much greater than what the poorest parts of the world are using, which is 1 gallon per day.

References

1. "Using Water Efficiently: Ideas for Industry." United States Environmental Protection Agency (EPA), Apr. 2000, URL: https://www.epa.gov/sites/production/files/2017-03/documents/ws-ideas-for-industry.pdf. Accessed 22 Feb. 2020.
2. "Water Conservation Tips for Businesses." EPA, URL: https://www3.epa.gov/region1/eco/drinkwater/water_conservation_biz.html. Accessed 23 Feb. 2020.
3. Erwin, Megan. "6 Steps to More Effective Water Conservation for Businesses." Pacific Gas and Energy Company (PG&E), URL: https://www.pge.com/en/mybusiness/save/smbblog/article/6-steps-to-more-effective-water-conservation-for-businesses.page. Accessed 23 Feb. 2020.
4. Crotta, Carol. "11 Ways to Save Water At Home." Forbes, 31 Mar. 2015, URL: https://www.forbes.com/sites/houzz/2015/03/31/11-ways-to-save-water-at-home/#5f950451166c. Accessed 23 Feb. 2020.

6 Question 6 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 2.5 pts** not enough components of ways to improve water use
- **- 5 pts** not enough components of ways to improve water use
- **- 7.5 pts** not enough components of ways to improve water use
- **- 2.5 pts** not enough components of record of water use and compare
- **- 5 pts** not enough components of record of water use and compare
- **- 15 pts** Question not answered